

Fantasy Fraud League

A SECURITY-BASED FANTASY GAME



Project Overview

Fantasy Fraud League (FFL) A security-based fantasy game where players run virtual banks, invest in fraud prevention tools, and react to fraud attacks launched by an admin. A fullstack project - with a frontend, backend, and database of your choosing. Think Fantasy Football/F1 League but with banks and security measures instead of teams and players.

Each player starts with \$1,000,000 in a virtual bank account. Players can buy up to x security tools to protect their bank from fraud. An admin user can trigger fraud attacks (random or selected), and players will gain or lose money depending on how well their defenses match the attack. The defense logic, money loss/gain, and pricing system is designed by the students.

Milestone 1: Research & Account Setup

Focus: Understand types of fraud and how to prevent them and set up authentication and user roles.

Requirements:

- Research 20 real-world fraud types (e.g., phishing, ATM skimming, deepfakes, synthetic identity fraud etc)
- Create a list of fraud prevention tools (e.g., encryption, 2FA, AI monitoring, CCTV), each with a made-up cost
- Each tool should map to at least one fraud type it protects against
- Design login and registration and store all user data (balance, tools selected) in a database

Milestone 2: Tool Shop & Team Management

Focus: Let users spend their budget to build their defense strategy.

Requirements:

- Players can choose up to x fraud prevention tools from the available list
- Display user's tools, balance etc
- Save all selections in the database

Milestone 3: Admin Attack Engine

Focus: Enable admin to trigger attacks and calculate player outcomes

Requirements:

- Admin dashboard must include:
 - A button to trigger a random fraud attack
 - A dropdown to select a specific fraud type
- When an attack is triggered: implement game logic (if protected/if not protected etc.. - update balance)
- Show the attack result to users (type of attack, impact, new balance)

Note: The system for how much money is gained/lost, and whether tools offer partial protection, can be simple or advanced — it's up to the student group.

Stretch Goal: Leaderboards & Leagues

Focus: Add multiplayer features and competitive insights

Optional Features:

- Allow users to create or join leagues
- Add a leaderboard showing player rankings by current balance
- Display basic stats like most chosen tool and best-performing user this week etc...

Key Learning Outcomes

- Gain an understanding of financial fraud and prevention techniques
- Fullstack development: login/auth, role-based access, API design, database usage
- Security best practices

References:

- F1 Fantasy league: <https://fantasy.formula1.com/en/create-team>
- Football fanatsy: <https://fantasy.premierleague.com>